

Web programming

Laboratórios de Sistemas e Serviços

Mário Antunes

2026

Universidade de Aveiro

JavaScript

JavaScript: Detailed Overview i

JavaScript (JS) is often misunderstood as a “scripting toy,” but it is a sophisticated, high-level language.

1. Dynamic Typing & Weak Typing

- Variables are not bound to a specific *data type*.
- *Why this matters:* You can assign a Number to a variable, and later assign a String to the same variable. This offers flexibility but increases the risk of runtime errors (e.g., trying to multiply a string).

```
let x = 42;  
x = "hello";  
console.log(x)
```

2. Prototype-based Object Orientation

- *How it works:* Unlike Class-based languages (Java/C++) where objects are instantiated from “blueprints” (classes), JS objects inherit directly from other objects (prototypes).
- *Implication:* Memory efficiency involves cloning existing structures rather than defining rigid hierarchies.

JavaScript: Detailed Overview iii

```
let person = {  
  eats: true,  
  hasLegs: 2,  
  walks(){ console.log('I can walk')}  
}  
//define another object  
let man = {  
  hasBreast: false,  
  hasBeard : true,  
}
```

JavaScript: Detailed Overview iv

```
//set the prototype of man to person object
man.__proto__ = person;
//define a third object
let samuel = {
  age: 23
}
//set the prototype of samuel to man
samuel.__proto__ = man;
//access walk method from samuel
console.log(samuel.walks())
//access hasBeard from samuel
console.log(samuel.hasBeard)
```

3. Single-Threaded Execution

- *The Constraint:* JS has a **single Call Stack**. It can only do *one thing at a time*.
- *The Risk:* If you run a heavy mathematical loop (e.g., calculating Pi to a billion digits), the entire browser tab freezes (UI blocking) because the thread is busy.

Programming Paradigms

Sequential Programming (Procedural)

This is the model used in basic C, Fortran, or simple Python scripts.

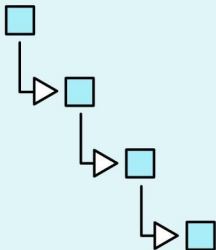
The Logic:

1. The program starts.
2. Line 1 executes.
3. Line 2 executes.
4. **Line 3 asks for input (scanf, input ()).**
5. The program **HALTS** (blocks) and waits for the user.
Nothing else happens until the user hits Enter.

Sequential Programming (Procedural) ii

Why this fails for UI: In a web interface, we cannot **"halt"** the rendering engine to wait for a mouse click. If we did, buttons wouldn't animate, and gifs wouldn't play.

SEQUENCES



SELECTIONS



LOOPS



Event-Driven Programming i

Modern interfaces (Web, Windows, macOS) use an **Event-Driven** architecture.

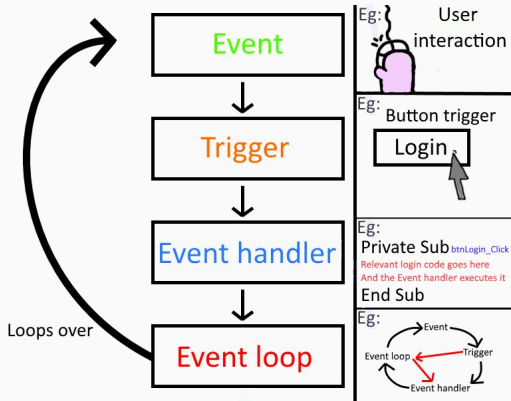
The Logic:

1. The program starts (Initialization).
2. It defines “Handlers” (functions waiting for specific triggers).
3. It enters the **Event Loop**.
4. The program sits in a **Listening State**.

The “Hollywood Principle”:

- *Don't call us, we'll call you.*
- The code doesn't ask “Did the user click?”. Instead, the browser interrupts the code saying “A click just happened, run the Click Function.”

Event-Driven Programming ii



How does a physical action become code execution?

1. **Hardware Level:** User moves the mouse. The mouse hardware sends an electrical signal (interrupt) to the CPU.
2. **OS Level:** The Operating System (Windows/Linux) interprets this signal as a coordinate change and paints the cursor moving.
3. **Browser Level:** The browser window sees the cursor is over a specific HTML button and the mouse button was pressed.

4. **The Event:** The browser creates a JavaScript Event Object containing details (X/Y coordinates, which button, timestamps).
5. **The Listener:** The browser checks: *Does this HTML element have a listener attached?*
6. **Execution:** If yes, the registered JS function is pushed to the execution stack.

JavaScript in the Webpage

The Document Object Model (DOM)

The Concept: When you write an HTML file, it is just a text string. The browser parses this string into a structure in memory called the DOM.

- **HTML:** `<div id="app"></div>` (Text on hard drive)
- **DOM:** `HTMLDivElement` (Object in RAM)

Why JS uses the DOM: JavaScript cannot edit the text file on the server. It edits the **Object in RAM**. The browser's rendering engine constantly watches the DOM; when JS updates the DOM object, the browser repaints the screen.

HTML is parsed sequentially (top to bottom). When the parser sees a `<script>` tag, it pauses HTML parsing to download and run the script. This creates issues:

1. The “Bottom of Body” Hack

- *Technique:* Putting `<script>` just before `</body>`.
- *Reasoning:* Ensures all HTML elements exist in the DOM before the script tries to find them.

2. The defer Attribute (Modern Standard)

```
<script src="app.js" defer></script>
```

- *Behavior:* The script downloads in the background (parallel) while HTML parses.
- *Execution:* The browser guarantees the script will only run **after** the HTML is fully parsed but **before** the DOMContentLoaded event.
- *Benefit:* Faster page load times and safe DOM access.

The Event Loop (Technical Detail)

How does single-threaded JS handle asynchronous tasks (like fetching data) without freezing?

1. **Call Stack:** Runs synchronous code (LIFO).
2. **Web APIs:** When you call `setTimeout` or `fetch`, the “work” is offloaded to the Browser’s C++ threads (not the JS thread).
3. **Callback Queue:** When the Web API is done, it puts your callback function into a Queue.
4. **The Loop:** The Event Loop checks: *“Is the Stack empty?”*
 - If **NO**: Wait.
 - If **YES**: Move the first item from the Queue to the Stack.

This is why `setTimeout(fn, 0)` doesn’t run immediately—it waits for the stack to clear.

JS Examples

1. Handling Mouse Events

We use `addEventListener`. This is the registration phase of Event-Driven programming.

```
const box = document.querySelector('#box');  
// The 'event' object is automatically passed by the browser  
function handleMove(event) {  
  // Update text with mouse coordinates  
  box.textContent = `X: ${event.clientX}, Y: ${event.clientY}`;  
  // Dynamic styling based on logic  
  if (event.clientX > 500) {  
    box.style.backgroundColor = 'red';  
  } else {  
    box.style.backgroundColor = 'blue';  
  }  
}  
// Subscribe to the 'mousemove' event  
box.addEventListener('mousemove', handleMove);
```

2. Dynamic Content (Photo Library)

We can create the interface programmatically. This is how React/Vue work under the hood (Imperative approach).

```
const urls = ['img1.jpg', 'img2.jpg'];
const container = document.getElementById('gallery');

urls.forEach(url => {
  // 1. Create Element: Creates an orphan object in memory
  const img = document.createElement('img');
  // 2. Configure Object: Set properties
  img.src = url;
  img.className = 'thumbnail';
  // 3. Attach Event: Make it interactive immediately
  img.addEventListener('click', () => {
    console.log("You clicked " + url);
  });
  // 4. Mount: Insert into the live DOM tree.
  container.appendChild(img);
});
```

3. Asynchronous Data (Fetch API)

Fetching data from an API takes time (latency). We use **Promises** (`async/await`) to prevent blocking.

```
async function getData() {  
  try {  
    // 'await' yields the thread until the Promise resolves.  
    // The UI remains responsive during this pause.  
    const response = await fetch('https://api.data.gov/users')(http);  
    // Parsing JSON is also asynchronous (it handles streams)  
    const data = await response.json();  
    console.log(data); // Runs only after network finishes  
  } catch (error) {  
    // Handles network failures (404, 500, Offline)  
    console.error("Fetch failed:", error);  
  }  
}
```

4. Real-Time Communication (WebSockets)

HTTP vs. WebSockets:

- **HTTP:** Client asks, Server answers, Connection closes. (Stateless).
- **WebSocket:** Client performs a “Handshake”, Connection upgrades to TCP socket, Connection stays open.

```
const socket = new WebSocket('ws://localhost:8080');  
// Event: Connection Established  
socket.onopen = () => {  
    console.log("Connected to Chat Server");  
    socket.send("User joined");  
};  
// Event: Server pushed data to us  
socket.onmessage = (event) => {  
    // This fires whenever the server sends data. No polling!  
    const message = JSON.parse(event.data);  
    displayMessage(message);  
};
```


Debugging in the Browser

The Challenge of Interpreted Languages i

Unlike C, C++, or Rust, JavaScript is an **Interpreted** (or JIT compiled) language.

Compiled Languages (C/C++):

- The compiler scans the entire code **before** execution.
- Syntax errors and type mismatches are caught at **Compile Time**.
- *Result:* You cannot run the program until these errors are fixed.

The Challenge of Interpreted Languages ii

Interpreted Languages (JavaScript):

- The browser reads and executes code line-by-line (or block-by-block) at **Runtime**.
- *Result:* The application might load perfectly and run for minutes.
- **The Crash:** The error only occurs when the execution flow hits the specific buggy line (e.g., when a user clicks a specific button).

Consequence: “It works on my machine” is common. You might not encounter the error because you didn’t trigger the specific execution path that contains the bug.

The Environment Gap: Editor vs. Browser i

Debugging Web Applications introduces a disconnect between where you **write** code and where you **run** code.

1. The Context Switch:

- You write code in an **IDE** (VS Code), which has static analysis and linting.
- You run code in the **Browser** (Chrome/Firefox).
- When an error occurs, it appears in the Browser's Console, not immediately in your text editor.

2. The “Black Box” Problem:

- The browser often runs “minified” or “bundled” code (to save bandwidth).
- An error on line 1 of `bundle.js` is useless to the developer.
- *Solution:* We rely on **Source Maps**, which tell the browser how to map the running code back to your original files.

1. "Printf" Debugging (`console.log`)

- The oldest method. You print variables to the browser console to inspect the state.
- *Pros*: Fast, simple.
- *Cons*: Clutters code, requires cleanup, doesn't pause execution.

2. The debugger ; Keyword

- Placing the statement `debugger` ; in your code forces the browser to **pause execution** (breakpoint) at that line.
- You can then step through code line-by-line.

3. Browser DevTools (The Sources Tab)

- Modern browsers (Chrome/Firefox) have built-in debuggers that rival desktop IDEs.
- You can set breakpoints, watch variables, and inspect the Call Stack directly in the browser.

Modern Frontend Frameworks

The “State vs. View” Problem

In complex apps (e.g., Facebook, Spotify), keeping the UI (View) in sync with the data (State) using Vanilla JS is error-prone.

Frameworks solve this by:

1. **Declarative Programming:** You define *what* the UI should look like for a given state, not *how* to update it.
2. **Componentization:** Breaking the UI into reusable, isolated chunks.

Developed by Facebook (Meta). React is technically a **Library**, not a Framework, focused solely on the View layer.

Key Concepts:

1. **Virtual DOM:** React keeps a lightweight copy of the DOM in memory. When state changes, it calculates the “diff” and updates only the changed parts of the real DOM.
2. **JSX (JavaScript XML):** Syntax extension allowing HTML to be written inside JS.
3. **Unidirectional Data Flow:** Data flows down (Parent -> Child).

React Example

Note the **Declarative** nature. We don't call `appendChild`. We return the structure we want.

```
import React, { useState } from 'react';
function ImageGallery() {
  // State Hook: When 'images' changes, the UI auto-updates
  const [images, setImages] = useState([
    { id: 1, url: 'img1.jpg' }
  ]);
  return (
    <div id="gallery">
      { /* Loop inside JSX */ }
      {images.map(img => (
        <img key={img.id} src={img.url}
          className="thumbnail" />
      ))}
    </div>
  );
}
```

Angular: The Framework

Developed by Google. Angular is a full-fledged **Framework**. It includes routing, HTTP clients, and form handling out of the box.

Key Concepts:

1. **TypeScript:** Mandatory. Adds static typing (Interfaces, Classes) to JS for safety.
2. **Dependency Injection (DI):** Built-in system to manage services and state.
3. **Two-Way Data Binding:** Changes in UI update State; Changes in State update UI (automatically).
4. **Real DOM:** Angular operates directly on the DOM but uses a sophisticated Change Detection mechanism (Zones).

Angular Example

Angular separates the Logic (Typescript) from the View (HTML Template).

Component Logic (gallery.component.ts)

```
import { Component } from '@angular/core';
@Component({
  selector: 'app-gallery',
  templateUrl: './gallery.component.html'
})
export class GalleryComponent {
  // Typed Array
  images: Array<{url: string}> = [{ url: 'img1.jpg' }];
}
```

Template (gallery.component.html)

```
<div id="gallery">
  <!-- Structural Directive (*ngFor) -->
  <img *ngFor="let img of images"
    [src]="img.url">
```

Summary Comparison

Feature	Vanilla JS	React	Angular
Paradigm	Imperative	Declarative	Declarative
Language	JavaScript	JS + JSX	TypeScript
DOM	Direct Access	Virtual DOM	Real DOM + Zones
Scale	Small scripts	Medium/Large Apps	Enterprise Apps
Learning Curve	Low	Medium	High

Backends

Node.js is not a language; it is a **Runtime Environment**. It takes Chrome's V8 Engine and adds C++ bindings for File System (FS) and Networking, allowing JS to run on servers.

NPM (Node Package Manager):

- Manages dependencies (libraries).
- **package.json**: The project manifest. Lists what libraries are needed (dependencies) and how to run the project (scripts).

Simple Express Server

Express is the standard framework for Node. It simplifies routing.

```
// Import express library
const express = require('express');
const cors = require('cors'); // Middleware for Security
const app = express();
// Enable CORS: Allows our browser-based JS (from a different origin)
// to fetch data from this server. Without this, the browser blocks it
app.use(cors());
// Define a Route (Endpoint)
app.get('/api/hello', (req, res) => {
  // Send JSON response
  res.json({
    msg: "Hello World",
    serverTime: Date.now()
  });
});
app.listen(3000, () => console.log("Running on port 3000"));
```

While Node.js shares a language with the frontend, **Python** is dominant in Data Science and AI.

FastAPI Features:

1. **Asynchronous:** Uses Python's `async def` (ASGI standard), making it much faster than Flask/Django.
2. **Type Hints:** Validates data automatically.
3. **Swagger UI:** Generates a documentation website (`/docs`) for your API automatically.

FastAPI Example

```
from fastapi import FastAPI
from fastapi.middleware.cors import CORSMiddleware

app = FastAPI()

# CORS Configuration
# Explicitly allowing the frontend container/origin
app.add_middleware(
    CORSMiddleware,
    allow_origins=["*"], # In prod, replace * with specific domain
    allow_methods=["*"],
)

@app.get("/api/items")
async def read_items():
    # Python Dictionary is automatically converted to JSON
    return [
        {"name": "Item 1", "price": 10.5},
        {"name": "Item 2", "price": 20.0}
    ]
```

We have two separate applications:

1. **Frontend:** Static HTML/JS served by Nginx (or a built React/Angular app).
2. **Backend:** Python/Node API processing data.

We need to run them together and ensure they can communicate.

Docker Compose Configuration

`docker-compose.yml` orchestrates multi-container applications.

```
services:
  # --- THE BACKEND ---
  backend-api:
    build: ./backend_folder           # Build image from Dockerfile
    container_name: py_api
    ports:
      - "8000:8000"                 # Expose port 8000 to host
    volumes:
      - ./backend_folder:/app       # Hot-reload code changes

  # --- THE FRONTEND ---
  frontend-web:
    image: nginx:alpine             # Use pre-built Nginx
    container_name: my_website
    ports:
      - "8080:80"                   # Browser hits localhost:8080
    volumes:
      # Inject our HTML/JS (or React build) into Nginx
```

Additional Resources

JavaScript & The Web * [MDN Web Docs \(Mozilla\)](#) - The bible of web development. * [JavaScript.info](#) - Deep dive into the modern language. * [What the heck is the event loop anyway?](#) (Philip Roberts) - Essential visualization of the JS runtime.

Frameworks * [React Documentation](#) - Official docs (newly rewritten). * [Angular University](#) - Comprehensive tutorials for Angular.

Backend & DevOps * [Node.js Best Practices](#) - Architecture patterns. * [FastAPI User Guide](#) - Excellent documentation with interactive examples. * [Docker Curriculum](#) - A hands-on guide for beginners.